Exp:10
Date:24/4/26

Internetworking with Routers in Cisco Packet Tracer.

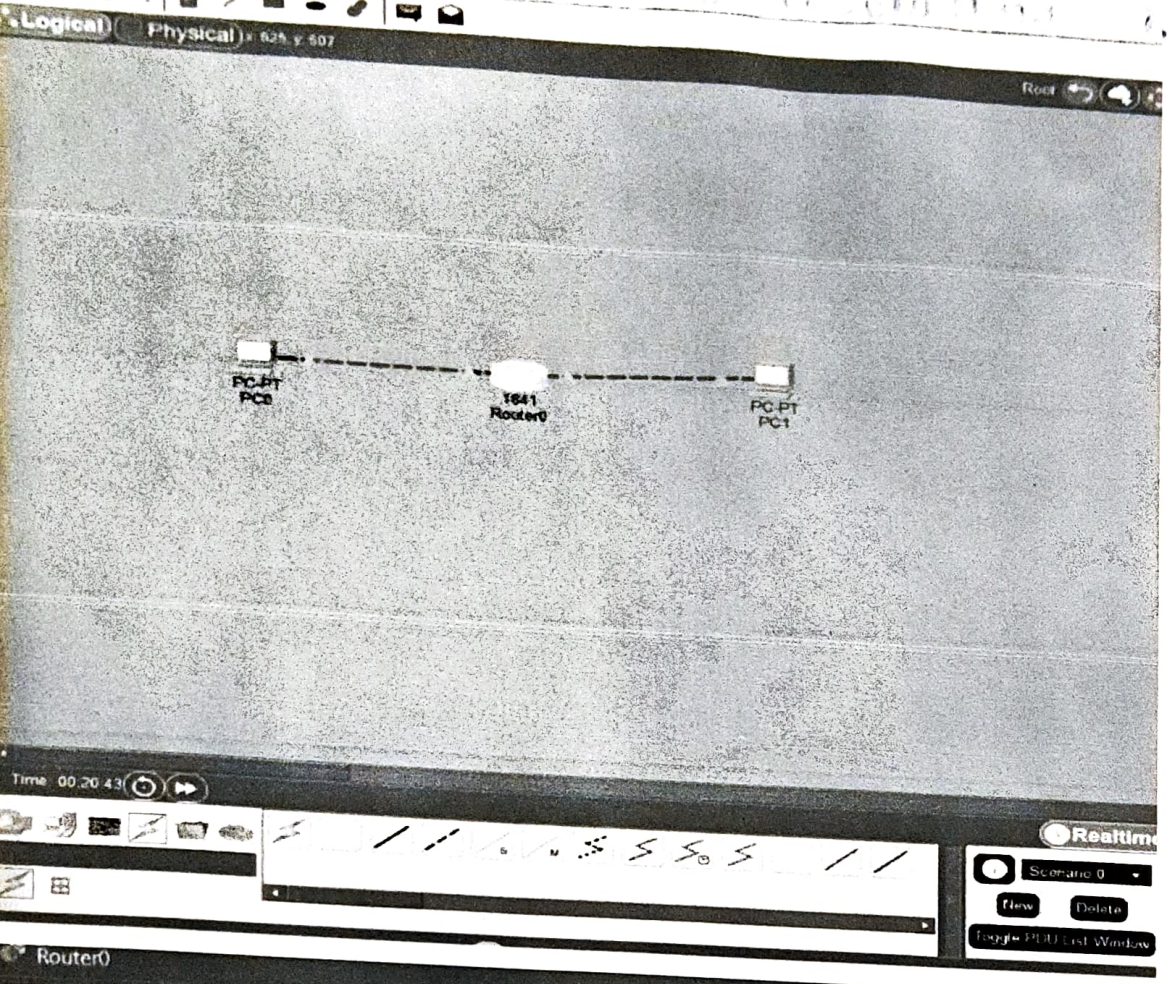
a) Design and Configure a Simple Internetwork Using a Router

Aim

To design and Configure an Internetwork Consisting of multiple networks Connected through a router and observe packet transmission b/w different networks

Algorithm/Procedure:

1. Open Cisco Packet Tracer.
2. place a router, Switches, using Copper Straight through Cables.
3. Connect the PCs to Switches using Copper Straight through Cables.
4. Connect each Switch to a Separate router Interface
5. Assign Static IP addresses and Subnet masking to all PCs
6. Configure the router interfaces with IP addresses belonging to their respective networks.
7. Enable the router interfaces using the no Shutdown Command.
8. Configure the default gateway on each PC using the IP address of its local router Interface.



Router0

Physical | Config | CLI | Attributes

```
Press RETURN to get started!


Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface FastEthernet0/0
Router(config-if)#ip address 192.168.10.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up
exit
Router(config)#interface FastEthernet0/1
Router(config-if)#ip address 192.168.20.1 255.255.255.0
Router(config-if)#no shutdown

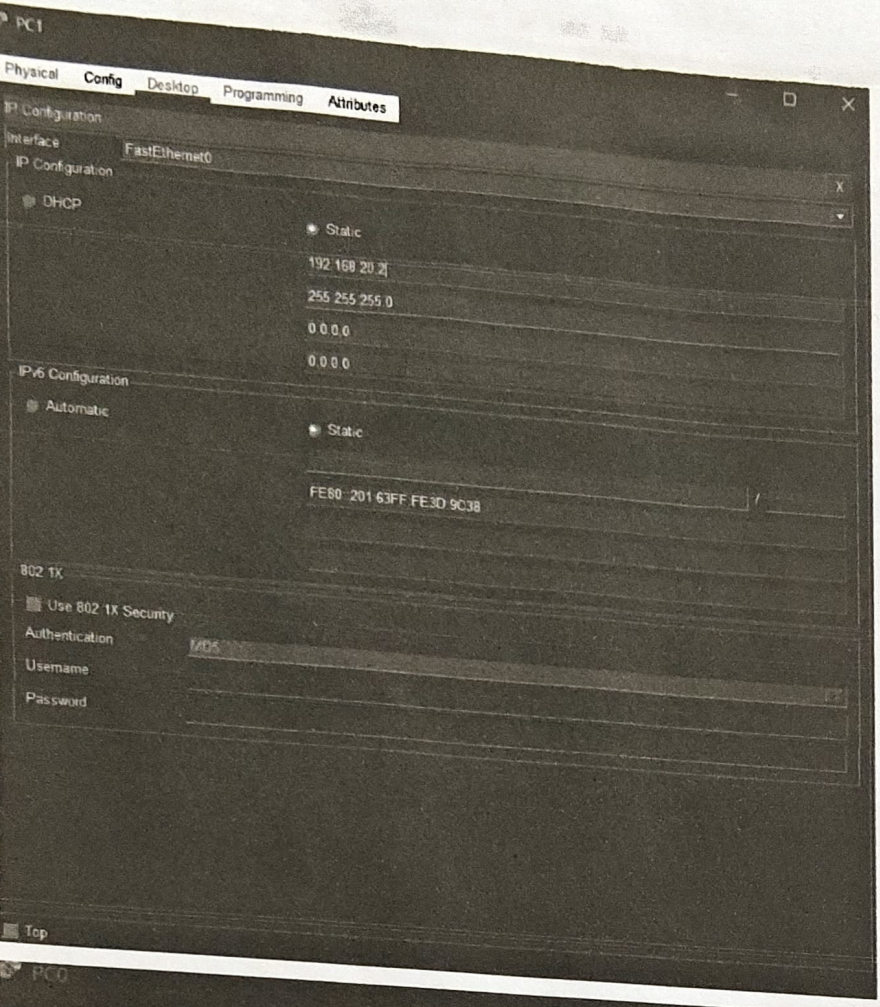
Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up
exit
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
end
Translating "end"...domain server (255.255.255.255)
% Unknown command or computer name, or unable to find computer address

Router#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
FastEthernet0/0        192.168.10.1    YES manual up                    down
FastEthernet0/1        192.168.20.1    YES manual up                    down
Vlan1                  unassigned      YES unset  administratively down down
Router#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up

Router#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
FastEthernet0/0        192.168.10.1    YES manual up                    up
FastEthernet0/1        192.168.20.1    YES manual up                    up
Vlan1                  unassigned      YES unset  administratively down down
Router#
```

Copy | Paste



PC0

Physical Config Desktop Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.20.2

Pinging 192.168.20.2 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.20.2:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
C:\>ping 192.168.20.2

Pinging 192.168.20.2 with 32 bytes of data:

Request timed out.
Reply from 192.168.20.2: bytes=32 time<1ms TTL=127
Reply from 192.168.20.2: bytes=32 time=1ms TTL=127
Reply from 192.168.20.2: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.20.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>
C:\>ping 192.168.20.2

Pinging 192.168.20.2 with 32 bytes of data:

Reply from 192.168.20.2: bytes=32 time<1ms TTL=127
Reply from 192.168.20.2: bytes=32 time<1ms TTL=127
Reply from 192.168.20.2: bytes=32 time<1ms TTL=127
Reply from 192.168.20.2: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.20.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

9. Test Connectivity b/w PCs in different networks using the Ping Command.

10. Use Simulation mode to observe how packets travel through the router.

Result:

A Simple Internetwork was designed and configured Successfully. Communication b/w different networks was verified, and packet forwarding through the router was observed.

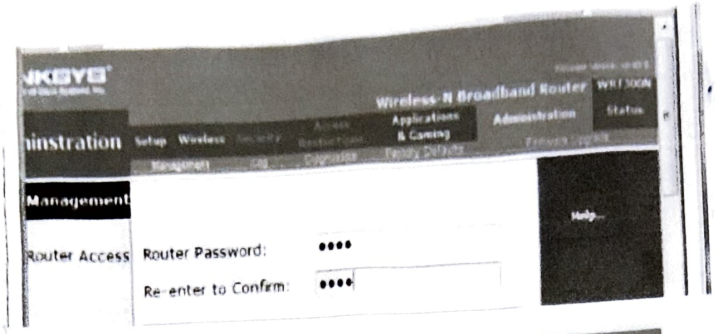
(b) Internetwork using Wireless Router, DHCP Server and Internet Cloud.

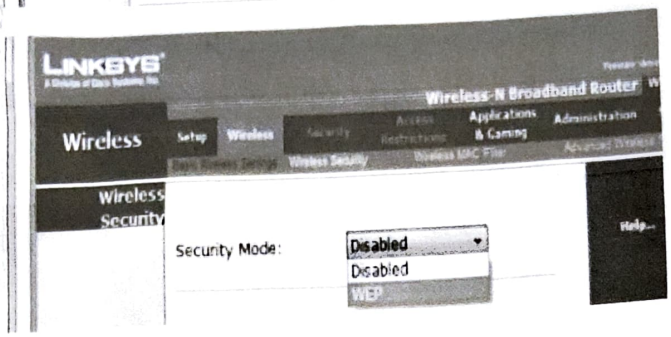
Aim:

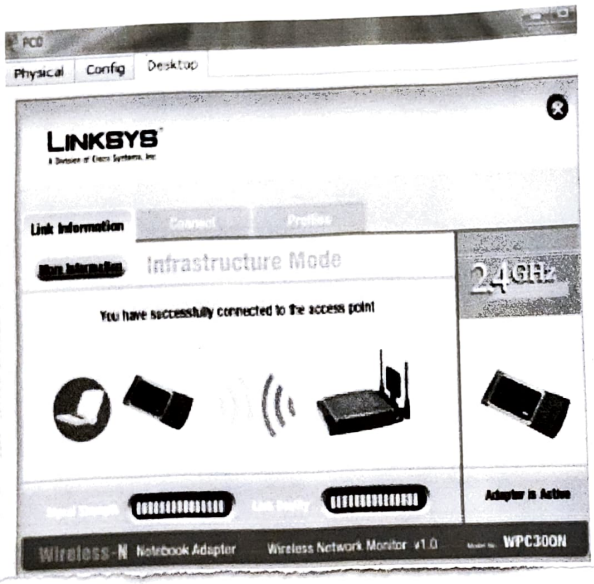
To configure an internetwork using a wireless router, DHCP server, and Internet cloud in Cisco Packet Tracer.

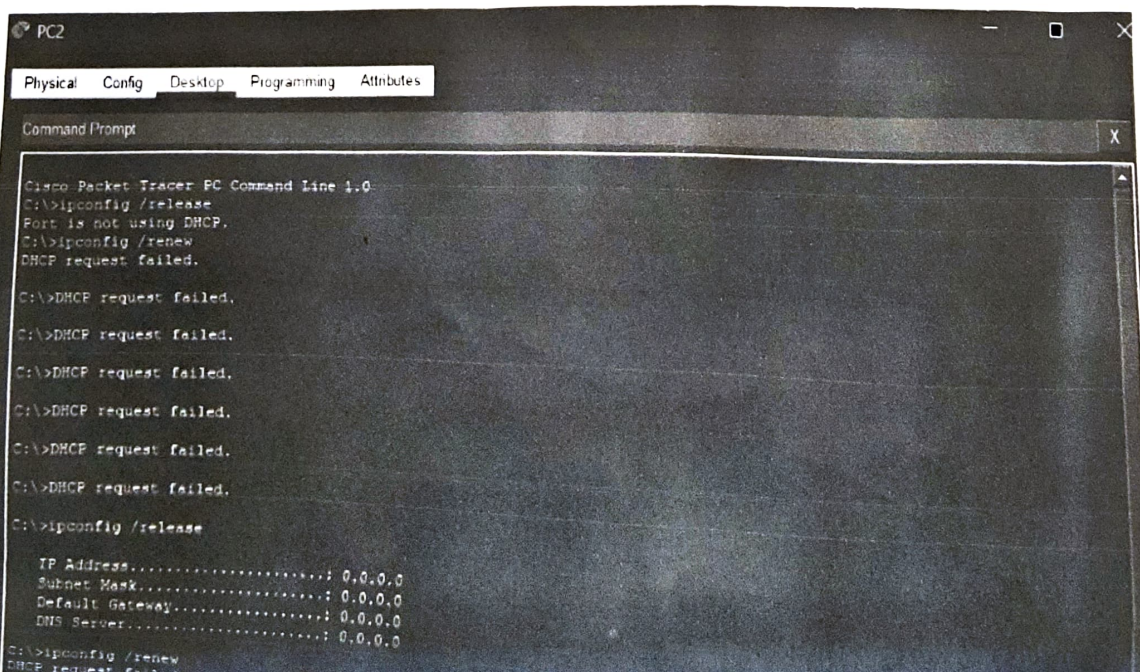
Algorithm/Procedure:

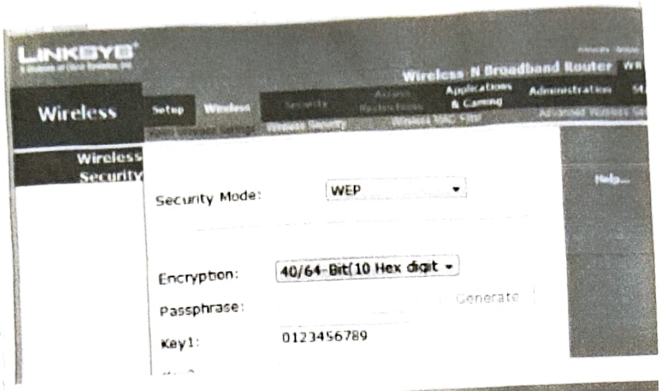
1. Open Cisco packet Tracer and place a wireless router, wireless end devices, a server, and an Internet cloud.
2. Configure the wireless router with an SSID and security settings.
3. Enable DHCP and configure the IP address pool, Subnet mask, default gateway, and DNS server if required.
4. Configure the server with a suitable static IP address and enable the required services.
5. Connect the router to the Internet cloud using the appropriate interface and connection.
6. Connect wireless devices to the configured SSID.
7. Verify that the devices receive IP addresses automatically through DHCP.

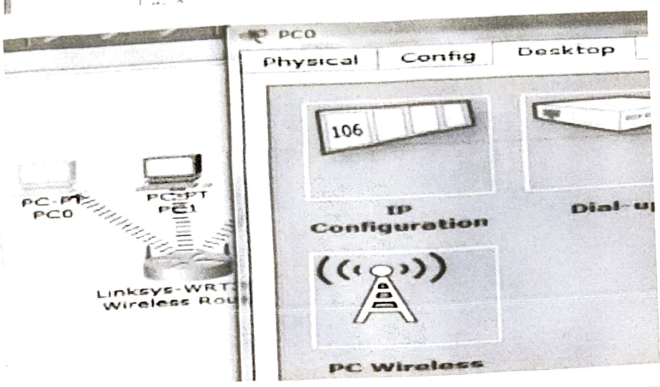

Wireless-N Broadband Router
Administration
Management
Router Access
Router Password:
Re-enter to Confirm:
Help...

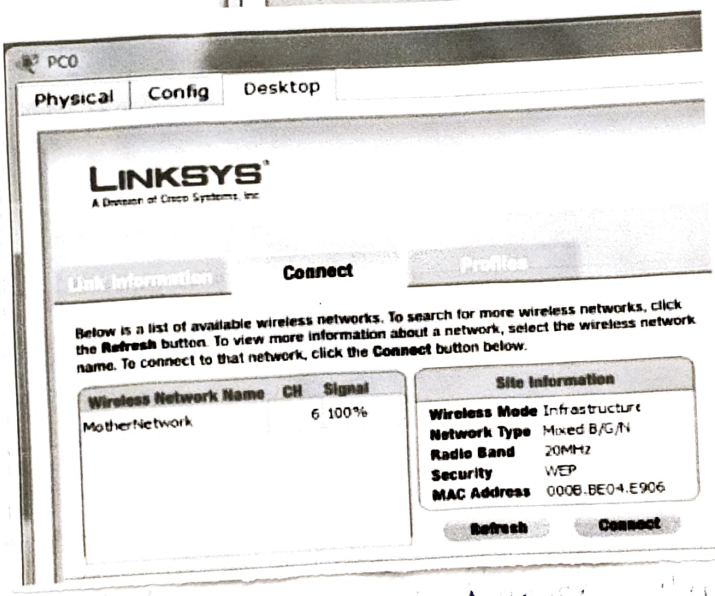

LINKSYS
Wireless-N Broadband Router
Wireless
Setup
Wireless
Applications & Gaming
Administration
Wireless Security
Security Mode:
Disabled
Disabled
WEP
Help...

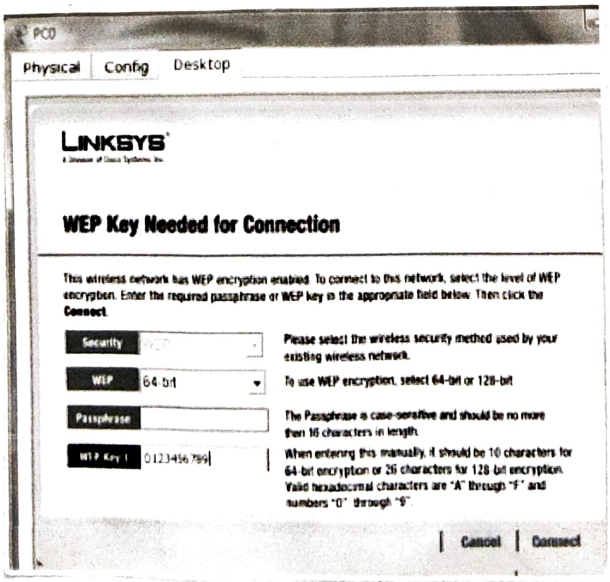

PC0
Physical
Config
Desktop
LINKSYS
Link Information
Infrastructure Mode
2.4GHz
You have successfully connected to the access point
Adapter is Active
Wireless-N Notebook Adapter
Wireless Network Monitor v1.0
WPC300N


PC2
Physical
Config
Desktop
Programming
Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ipconfig /release
Port is not using DHCP.
C:\>ipconfig /renew
DHCP request failed.
C:\>DHCP request failed.
C:\>DHCP request failed.
C:\>DHCP request failed.
C:\>DHCP request failed.
C:\>DHCP request failed.
C:\>DHCP request failed.
C:\>ipconfig /release
IP Address.....: 0.0.0.0
Subnet Mask.....: 0.0.0.0
Default Gateway.....: 0.0.0.0
DNS Server.....: 0.0.0.0
C:\>ipconfig /renew

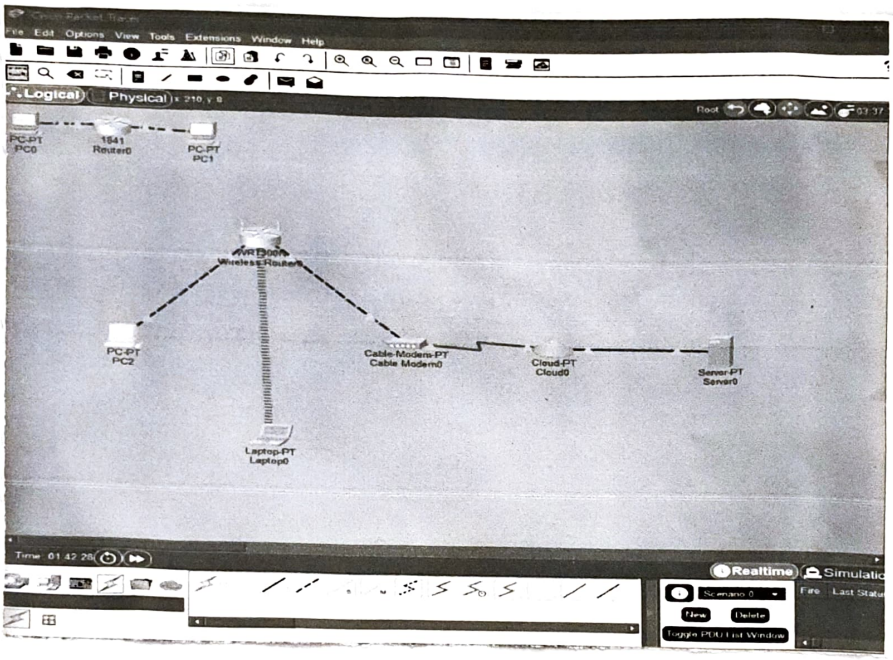

LINKSYS
Wireless-N Broadband Router
Wireless
Setup
Wireless
Security
Applications & Gaming
Administration
Wireless Security
Help...
Security Mode:
WEP
Encryption:
40/64-Bit(10 Hex digit
Passphrase:
Generate
Key1:
0123456789


PC0
Physical
Config
Desktop
106
IP Configuration
Dial-up
PC Wireless
PC-PT PC0
PC-PT PC1
Linksys-WRT
Wireless Rou


PC0
Physical
Config
Desktop
LINKSYS
A Division of Cisco Systems, Inc
Link Information
Connect
Profiles
Below is a list of available wireless networks. To search for more wireless networks, click the Refresh button. To view more information about a network, select the wireless network name. To connect to that network, click the Connect button below.
Wireless Network Name
CH
Signal
MotherNetwork
6
100%
Site Information
Wireless Mode Infrastructure
Network Type Mixed B/G/N
Radio Band 20MHz
Security WEP
MAC Address 0008.BE04.E906
Refresh
Connect

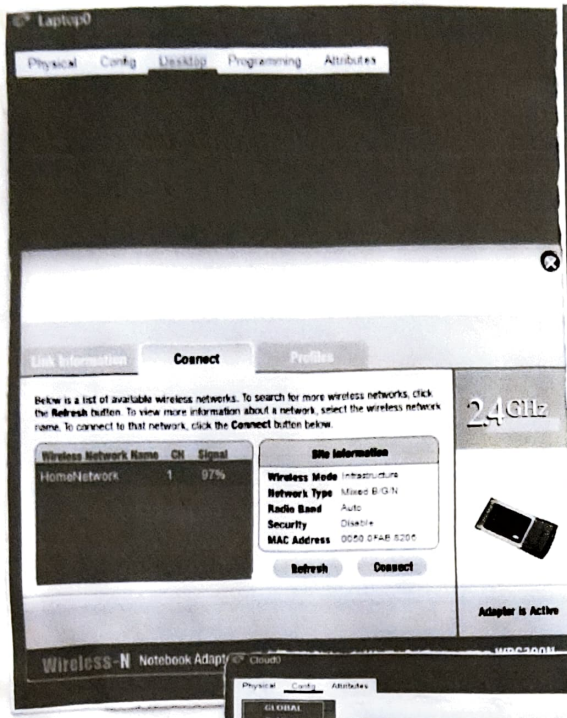

PC0
Physical
Config
Desktop
LINKSYS
WEP Key Needed for Connection
This wireless network has WEP encryption enabled. To connect to this network, select the level of WEP encryption. Enter the required passphrase or WEP key in the appropriate field below. Then click the Connect.
Security
WEP
Please select the wireless security method used by your existing wireless network.
WEP
64 bit
To use WEP encryption, select 64-bit or 128-bit
Passphrase
The Passphrase is case-sensitive and should be no more than 16 characters in length.
WEP Key 1
0123456789
When entering this manually, it should be 10 characters for 64-bit encryption or 26 characters for 128-bit encryption. Valid hexadecimal characters are "A" through "F" and numbers "0" through "9".
Cancel
Connect

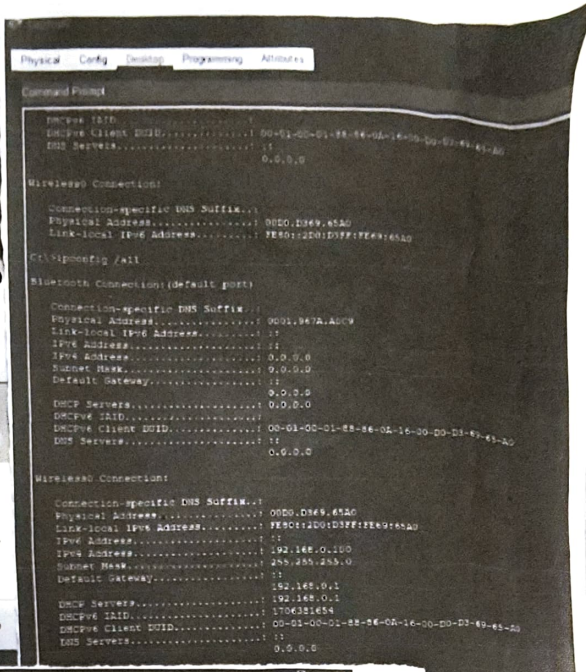
8. Test Connectivity to the server and other reachable network devices.

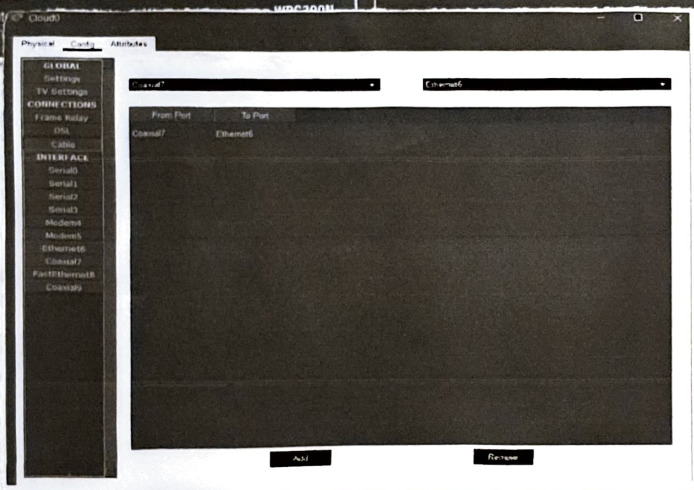


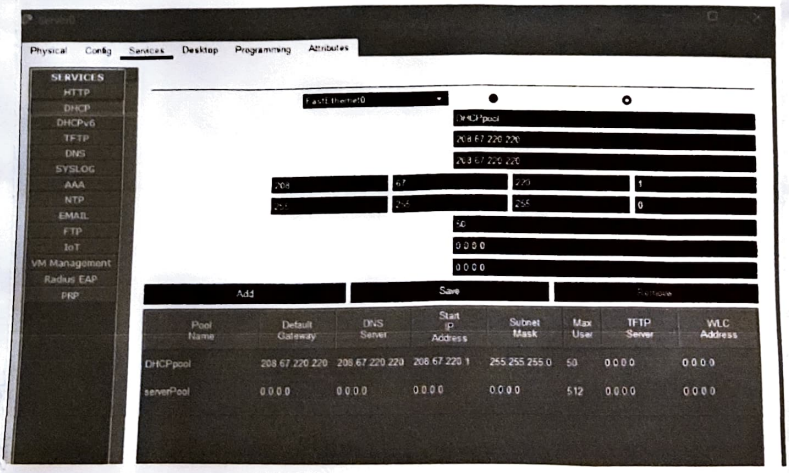
Result

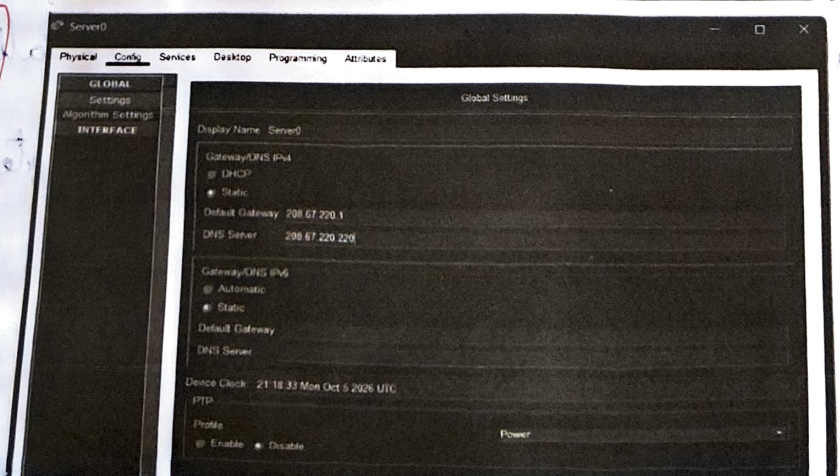
The wireless network was configured successfully. Devices obtained IP addresses automatically through DHCP, and connectivity was tested.


Laptop0
Physical
Config
Desktop
Programming
Attributes
Connect
Below is a list of available wireless networks. To search for more wireless networks, click the Refresh button. To view more information about a network, select the wireless network name. To connect to that network, click the Connect button below.
Wireless Network Name
CH
Signal
HomeNetwork
1
97%
Site Information
Wireless Mode
Network Type
Radio Band
Security
MAC Address
Refresh
Connect
2.4GHz
Adapter is Active
Wireless-N Notebook Adapt


Physical
Config
Desktop
Programming
Attributes
Command Prompt
Wireless0 Connection:
Physical Address.....: 0060.D369.65A0
Link-local IPv6 Address.....: FE80::2D0:D3FF:FE69:65A0
C:\>ipconfig /all
Bluetooth Connection:(default port)
Wireless0 Connection:
Physical Address.....: 0060.D369.65A0
Link-local IPv6 Address.....: FE80::2D0:D3FF:FE69:65A0
IPv4 Address.....: 192.168.0.100
Subnet Mask.....: 255.255.255.0
Default Gateway.....: 192.168.0.1
DNS Servers.....: 192.168.0.1


Cloud0
Physical
Config
Attributes
GLOBAL
Settings
TV Settings
CONNECTIONS
Frame Relay
DSL
Cable
INTERFACE
Serial0
Serial1
Serial2
Serial3
Modem4
Modem5
Ethernet6
Coaxial7
FastEthernet8
Coaxial9
From Port
To Port
Coaxial7
Ethernet6
Add
Remove


Physical
Config
Services
Desktop
Programming
Attributes
SERVICES
HTTP
DHCP
DHCPv6
TFTP
DNS
SYSLOG
AAA
NTP
EMAIL
FTP
IoT
VM Management
Radius EAP
PPP
FastEthernet0
DHCPpool
208.67.220.220
208.67.220.220
208
67
220
1
255
254
255
0
50
0.0.0.0
0.0.0.0
Add
Save
Remove
Pool Name
Default Gateway
DNS Server
Start IP Address
Subnet Mask
Max User
TFTP Server
WLC Address
DHCPpool
208.67.220.220
208.67.220.220
208.67.220.1
255.255.255.0
50
0.0.0.0
0.0.0.0
serverPool
0.0.0.0
0.0.0.0
0.0.0.0
0.0.0.0
512
0.0.0.0
0.0.0.0


Server0
Physical
Config
Services
Desktop
Programming
Attributes
GLOBAL
Settings
Algorithm Settings
INTERFACE
Global Settings
Display Name: Server0
Gateway/DNS IPv4
DHCP
Static
Default Gateway: 208.67.220.1
DNS Server: 208.67.220.220
Gateway/DNS IPv6
Automatic
Static
Default Gateway
DNS Server
Device Clock: 21:18:33 Mon Oct 5 2026 UTC
FTP
Profile
Enable
Disable
Power

(C) Internetwork Using Switches, a router, and Ethernet cables

Aim:

To design and configure an internetwork using switches, a router, and Ethernet cables in the laboratory.

Algorithm/procedure:

1. Place the required PCs, switches and router in Cisco Packet Tracer or arrange the devices in the laboratory.
2. Connect PCs to switches using Ethernet cables.
3. Connect the switches to the router using suitable Ethernet cables.
4. Assign static IP addresses and subnet masks to all PCs.
5. Configure the router interface with appropriate IP addresses.
6. Enable the router interfaces.
7. Configure the default gateway on each PC.
8. Test communication between devices in the same and different networks using ping.
9. Observe the packet transmissions and verify connectivity.

Result:

The internetwork was configured successfully using switches, a router, and Ethernet cables. Connectivity b/w the configured networks was verified.